Randomized and low-rank approximation

Nonadaptive queries for finite-family matrix approximation

Difficulty: challenging **Importance:** interesting to the community

Status: Lean verified

Rating rationale: Challenging because all query choices must precede information about the target; community impact is parallel and pass-efficient structured matrix learning.

Area: nonadaptive matrix sketching

Resolution — affirmative, 13 September 2026

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation) gives a complete affirmative answer in Theorem 1.1 and Sections 2–6 (TeX source). The algorithm fixes all query vectors and query sides before any answers, succeeds with probability at least 0.99 for every fixed input, and uses at most $4{,}000{,}000\sqrt{\log(2M)}\,\varepsilon^{-2}$ queries. Thus the original target holds with $C = 4{,}000{,}000$ and $b = 0$, including zero optimal residual and rank-deficient cases, in precisely the stated exact-real, query-only model.

The complete proof passed a separate independent Codex AI-agent informal audit, satisfying the repository's *Solved* policy. That audit was the evidence for the September 2026 status. Authorship, verified affiliation, original submission and checks. The original target and historical evidence below are retained; the stronger nonadaptive relative-error question is not claimed settled.

Lean proof and verification evidence

Lean verified. The formalization proves the original nonadaptive matrix-query guarantee with $C = 4{,}000{,}000$ and $b = 0$. The fixed query program has a $3 + \varepsilon$ error factor with probability at least $99/100$ under the Gaussian stream law. The proof covers zero optimal residual and rank-deficient matrices.

The independent reviews and isolated Linux checks cover the full original target in Lean 4.33.1. The proof uses the standard three axioms. Immutable proof and dependency pins · Reviews, reproduction, and verification logs.

Context and notation

Use exact real arithmetic, with comparisons, standard Gaussian sampling, and exact SVDs available as primitives; charge an SVD of an $a \times b$ matrix $O(ab\min(a,b))$ operations. This states the idealized arithmetic model used here, rather than a finite precision or bit complexity claim. A matrix–vector query returns either Av or $A^{\mathsf{T}}v$ for one chosen real vector v; both types count toward the total. Randomized guarantees are for every fixed input, with probability or expectation over the algorithm's randomness.

Problem statement

Let $n \geq 1$, let $\mathcal{F} \subset \mathbb{R}^{n\times n}$ be explicitly given with $M = |\mathcal{F}| \geq 2$, and let $A \in \mathbb{R}^{n\times n}$ be accessible only through matrix–vector queries. Set $t = \log(2M)$.

Question

Do absolute constants $C > 0$, integer $b \geq 0$, and a uniform randomized algorithm exist that, for every $0 < \varepsilon < 1/2$, choose all query vectors and all choices between A and A^{T} before receiving any oracle answers, make at most

$$C\sqrt{t}\,\varepsilon^{-2}[1 + \log(2 + t) + \log(1/\varepsilon)]^b$$

queries, and return $B \in \mathcal{F}$ such that

$$\Pr\left[\|A - B\|_F \leq (3 + \varepsilon)\min_{D\in\mathcal{F}}\|A - D\|_F\right] \geq 0.99?$$

Query choices may depend on $\mathcal{F}, \varepsilon$ and randomness. Only oracle calls are charged; candidate processing and processing of the answers are unrestricted.

References

Amsel et al., *Query Efficient Structured Matrix Learning*, §5, adaptivity question. Christopher Musco, *Structured Matrix Learning from Matrix-Vector Products*, ICERM, February 4, 2026, numbered slide 23.

Status evidence

Both sources explicitly ask whether adaptivity is necessary; the paper's August 21, 2026 revision retains the question. Searches for "finite matrix approximation adaptivity 2026" and the paper identifier with "non-adaptive" found no resolution. The abstract's linked finite-family improvement, Theorem 1.2, uses two rounds after receiving a constant-factor warm start: its left query vectors depend on the first round's answers. It does not supply the required nonadaptive algorithm.

Audit — 2026-09-10

Rechecked Amsel et al., §5 and the COLT 2026 record. The adaptivity question remains. Nonadaptive structured-learning searches found no resolution; the linked finite-family relative-error improvement still uses responses to choose later queries.